Question 014 | Question bank page 2 | Q.2

Develop an ASP.NET Core Web API to perform Create, Read, Update, and Delete operations on student records stored in a database using Entity Framework Core.

Project type: ASP.NET Core Web API

Steps to perform

1. In Visual Studio create an ASP.NET Core Web API project.

2. Add Models and Controllers files; configure required services in Program.cs.

3. If EF Core is used, install matching provider packages, set the connection string and apply migrations.

4. Build and run the application.

5. Test GET/POST/PUT/DELETE endpoints using Postman or Swagger where configured.

Closest supplied practical: } } Develop CRUD operations using ASP.NET Core Web API and Entity Framework Core.

Source: Dot_Net_Practicals_6_to_10_Organized_v2.pdf

Provided creation instructions and source files

```
Creation steps: Create the StudentAPIWithEF Web API project. Add EF Core packages, model, DbContext,
connection string and controller; configure services, then create the database with migrations.
Filename: Program.cs    Location: Program.cs
using Microsoft.EntityFrameworkCore;
using StudentAflIWithEF.Data;
var builder = WebApplication.CreateBuilder(args);
builder.Services.AddControllers();
builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(builder.Configuration.GetConnectionString("DefaultConnection")));
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();
var app = builder.Build();
if (app.Environment.IsDevelopment())
{
    app.UseSwagger();
    app.UseSwaggerUI();
}
app.UseHttpsRedirection();
app.UseAuthorization();
app.MapControllers();
app.Run();
Filename: Student.cs    Location: Models/Student.cs
namespace StudentAflIWithEF.Models
{
    public class Student
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public int Age { get; set; }
        public string Course { get; set; }
    }
}
Filename: StudentController.cs    Location: Controllers/StudentsController.cs
using Microsoft.AspNetCore.Mvc;
using StudentAflIWithEF.Data;
using StudentAflIWithEF.Models;
namespace StudentAflIWithEF.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentsController : ControllerBase
    {
        private readonly AppDbContext _context;
        public StudentsController(AppDbContext context)
        {
            _context = context;
        }
        [HttpGet]
        public IActionResult GetStudents()
        {
            return Ok(_context.Students.ToList());
        }
        [HttpGet("{id}")]
        public IActionResult GetStudent(int id)
        {
```

```
            var student = _context.Students.Find(id);
            if (student == null) return NotFound();
            return Ok(student);
        }
        [HttpPost]
        public IActionResult AddStudent(Student student)
        {
            _context.Students.Add(student);
            _context.SaveChanges();
            return Ok(student);
        }
        [HttpPut("{id}")]
        public IActionResult UpdateStudent(int id, Student student)

        {
            var data = _context.Students.Find(id);
            if (data == null) return NotFound();
            data.Name = student.Name;
         data.Age = student.Age;
            data.Course = student.Course;
            _context.SaveChanges();
            return Ok(data);
        }
        [HttpDelete("{id}")]
        public IActionResult DeleteStudent(int id)
        {
            var student = _context.Students.Find(id);
            if (student == null) return NotFound();
            _context.Students.Remove(student);
            _context.SaveChanges();
            return Ok();
        }
    }
}
Filename: Appcontext    Location: Data/AppDbContext.cs
using Microsoft.EntityFrameworkCore;
using StudentAflIWithEF.Models;
namespace StudentAflIWithEF.Data
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) { }
        public DbSet<Student> Students { get; set; }
    }
}
Filename: Apsetting.json/repos    Location: appsettings.json
{
  "ConnectionStrings": {
    "DefaultConnection":
"Server=(localdb)\\MSSQLLocalDB;Database=StudentDB;Trusted_Connection=True;TrustServerCertificate=True;"
  },
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*"
}
```

CHECK: This is a related source example, not necessarily a complete or exact solution to the wording above. Adapt and verify it.

Expected result / verification

The application should meet the exact question requirements. Record actual output or screenshots after running; no execution has been claimed in this manual.